Lesson 2: Understanding Limits Graphically and Numerically

Topic 1.2: Defining Limits and Using Limit Notation

Limits are the "backbone" of understanding that connect algebra and geometry to the mathematics of calculus. In basic terms, a limit is just a statement that tells you what height a function *INTENDS TO REACH* as you get close to a specific x-value. Recall from Pre-Calculus that you evaluated three types of limits. Complete the table below:

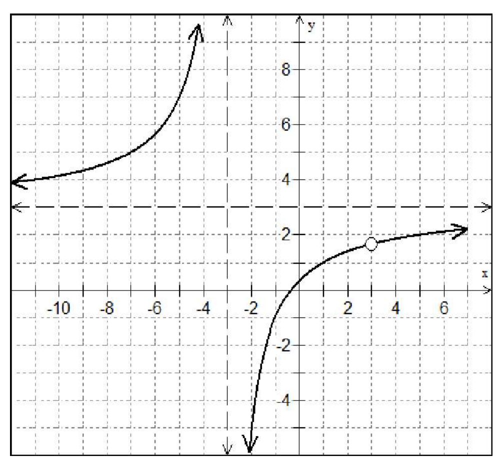
PROPER LIMIT NOTATIONS		
TYPE OF LIMIT	**PROPER NOTATION**	**VERBALLY:**
Right-hand limit		
Left-hand limit		
General limit		

Let's begin our discussion of limits by analyzing a rational function and examining the graph.

EX #1: Use the equation for $f(x)$ and the graph of the function to analyze completely.

$$f(x) = \frac{3x^2 - 8x - 3}{x^2 - 9}$$

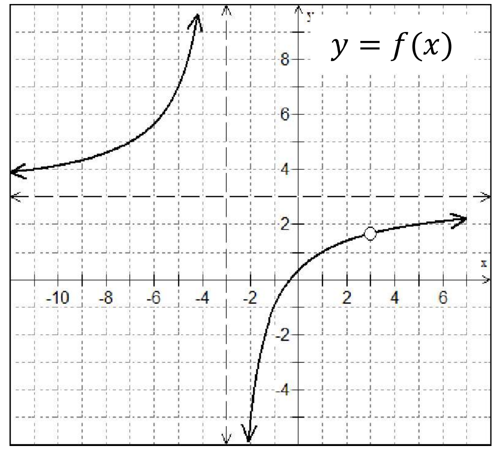
Factor and Simplify	
Coordinates of Hole	
Domain	
Range	
Vertical Asymptote	
Horizontal Asymptote	



Understanding Limit Notation

Let's revisit the notation we learned in Pre-Calculus. We need to convert these old methods of explaining extreme behaviors into limit notation for use in calculus.

EX #2: Use the graph to complete the table below.



Pre-Calculus Notation vs. Calculus Notation	
As $x \to -\infty$, the graph of $f(x) \to$ ________	As $x \to \infty$, the graph of $f(x) \to$ ________
As $x \to -3$ from the right, the graph of $f(x) \to$ ______	As $x \to 3$ from the left, the graph of $f(x) \to$ ________
As $x \to -3$ from the left, the graph of $f(x) \to$_______	As $x \to 3$ from the right, the graph of $f(x) \to$______

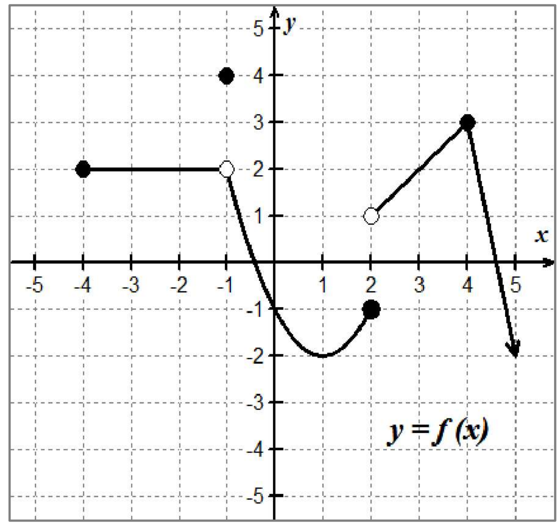
Can you explain what the value of a limit represents in terms of the graph?

Informally, a **<u>limit is a</u>** <u>y-value</u> which a function approaches as x approaches some value.

$\lim_{x \to c} f(x) = L$ means as x approaches c, $f(x)$ approaches the y-value of L.

Topic 1.3: Estimating Limit Values from Graphs

Consider the function shown at right.



Say you want to find $\lim_{x \to 4^+} f(x)$, the positive sign in the limit notation indicates a right-hand limit. If you think of the function as a highway and imagine you are traveling along the graph of $f(x)$ toward $x = 4$ **FROM THE RIGHT, NOT TO THE RIGHT**, and you stop at the vertical line $x = 4$, the y-value where you stop is 3. Therefore, $\lim_{x \to 4^+} f(x) = 3$.

EX #3: Use the graph above to evaluate each of the following limits:

A. $f(2)$	**B.** $f(-1)$
C. $\lim_{x \to 4^-} f(x)$	**D.** $\lim_{x \to 2^+} f(x)$
E. $\lim_{x \to 2^-} f(x)$	**F.** $\lim_{x \to -1^+} f(x)$
G. $\lim_{x \to -1^-} f(x)$	**H.** $\lim_{x \to -4^+} f(x)$
I. $\lim_{x \to -4^-} f(x)$	**J.** $\lim_{x \to -1} f(x)$
K. $\lim_{x \to 2} f(x)$	**L.** $\lim_{x \to 5} f(x)$
M. $\lim_{x \to 0} f(x)$	**N.** $\lim_{x \to 1} f(x)$

Think about this!

If we think of the function as a highway, then the point at $(2, -1)$ could be considered the end of the road, while the point at $(-1, 2)$ is more like a "pothole." How would you describe the points located at

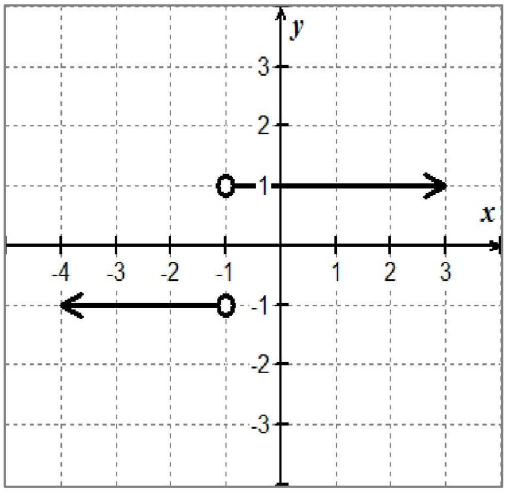
$(2, 1)$: ____

$(4, 3)$: ____

Hopefully, this analogy gives you a visual reference for understanding limits from a graphical approach. Let's get a little more formal with our definition now.

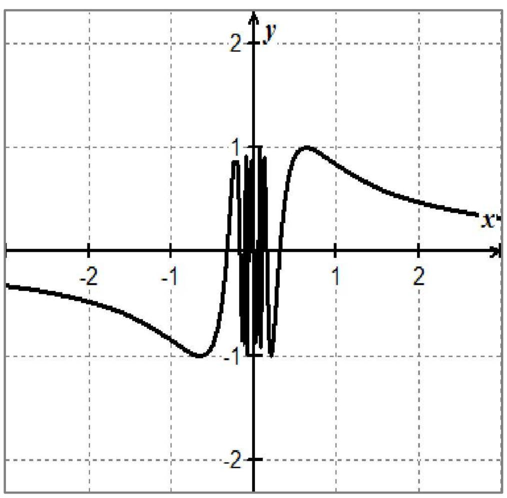
EX #4: Limits can fail to exist in three situations:

CASE 1: ____



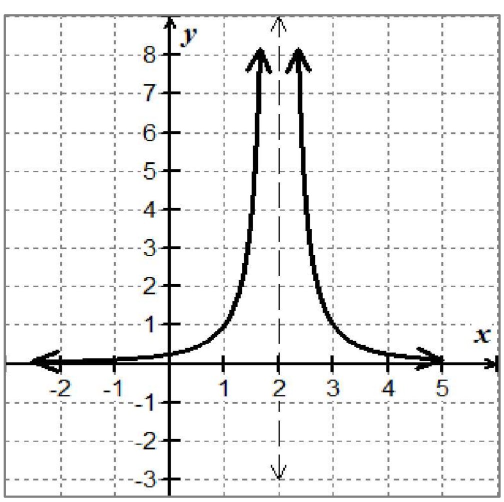
Justify why the limit does not exist at $x = -1$ for $f(x) = \frac{|x+1|}{x+1}$

CASE 2: ____



Justify why the limit does not exist at $x = 0$ for $f(x) = \sin\left(\frac{1}{x}\right)$

CASE 3: ____



Justify why the limit does not exist at $x = 2$ for $f(x) = \frac{1}{(x-2)^2}$

Topic 1.4: Estimating Limit Values from Tables

EX #5: Now, consider the function $f(x) = \frac{x-3}{x^2+2x-15}$. Complete the table below to find the limit as $x \to 3$.

$\boldsymbol{x}$	2.9	2.99	2.999	3	3.001	3.01	3.1
$\boldsymbol{f(x)}$							

Based on your analysis, what are the values of each of the limits below?

$\lim_{x \to 3^-} f(x) =$	$\lim_{x \to 3^+} f(x) =$	$\lim_{x \to 3} f(x) =$

By evaluating certain types of functions at a particular value, we may not necessarily have a sufficient understanding of the function's behavior at a specific point. This is especially true for rational functions that contain discontinuities.

This is why the idea of a ***limit is so important!*** Rather than just going directly to some x-value using direct substitution , we can **approach a point from either side** to get some sense of behavior in the **neighborhood**.

Look at the photo of the arch support construction for the Hoover Dam Bypass Bridge (2009). We can tell where the missing section is going to be. The actual height at that point would be the limit. With limits, we can discuss behavior at a point whether the point exists or not. With limits, we are looking at the y-value the graph approaches, not the actual y-value at that point.

When finding limits, ask yourself, "What is happening to y as x gets close to a certain number?" You are finding the ***y*-value** for which the function is approaching as x approaches c.

LIMIT EXISTENCE THEOREM:

$$\lim_{x \to c} f(x) \text{ exists if and only if } \lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L$$

where L is a real number.

Verbally: The limit as x approaches c on $f(x)$ will exist if and only if the limit as x approaches c from the left is equal to the limit as x approaches c from the right.

EX #6: YOU OWN IT! In the box below, complete the sentence in your own words.

In order for the GENERAL LIMIT to exist, the function:

EX #7: Sketch a graph to satisfy each set of conditions.

1. $f(a)$ is undefined
2. $x = a$ is a point discontinuity
3. $\lim_{x \to a} f(x)$ exists

1. $\lim_{x \to a} f(x)$ DNE
2. $x = a$ is a jump discontinuity
3. $f(a)$ is defined